



Mohammed HSINY

Electrical Engineering & Industrial Control Graduate Student

CONTACT

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📍 Safi, Morocco
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📅 May 4, 2003

EDUCATION

Engineering Degree – Electrical Engineering & Industrial Control
FST Mohammedia 2024–2027
BSc in Science and Technology – Industrial IT & Instrumentation
FST Errachidia 2023–2024
DEUST – Mathematics, Computer Science & Physics
FST Errachidia 2021–2023
Baccalaureate in Physical Sciences
Lycée Najib Mahfoud, Safi 2021

SKILLS

Embedded Systems, Robotics & AI
Embedded programming, communication protocols, IoT, Real-time systems, Machine Learning, Image processing, Advanced control laws, Autonomous systems, Navigation & guidance.
Automation & Industrial Control
PLC, SCADA, Instrumentation, Industrial networks, Industrial automation & maintenance, Electrical installations (CFO/CFA), Power electronics, Embedded power supply.
Mechanical Design & Hands-on
Mechanical design, 3D printing, Mechanical & bolted assembly, Troubleshooting, Soldering & wiring.

TOOLS

Embedded, Robotics & AI
STM32 / STM32CubeIDE, Arduino / IDE, Raspberry Pi, ESP32, Betaflight, ArduPilot, INAV, QGroundControl, MAVLink, Mission Planner, ROS, KiCad, VS Code.
Automation & Industrial Control
Schneider Control Expert, Siemens STEP 7, TIA Portal, GMAO, CANECO BT/HT, AutoCAD, NF C 15-100.
Development & AI
Python, C/C++, OpenCV, YOLO, Git, Pascal, LLMs (Codex, Claude).
CAD, Simulation & Fabrication
SolidWorks, MATLAB, Gazebo, Factory.

LANGUAGES

Arabic : Native
French : Professional proficiency (C1)
English : Professional proficiency (B2)

AWARDS

- 1st Prize: International Summer School ENSEM | ENSAO | FSTM | FSBM Casablanca.
- 2nd Prize: ENSAKH | ENIMR.
- 3rd Prize: EMI Rabat | ENSA El Jadida.

PROFESSIONAL PROFILE

Electrical Engineering & Industrial Control graduate student, seeking a PFE apprenticeship in the fields of **embedded systems, robotics, UAVs or autonomous vehicles for defense**. Passionate about defense technologies and critical embedded systems, I actively follow the latest developments in the drone and autonomous systems market. Curious, self-driven and possessing strong **initiative**, I develop personal projects integrating both hardware and software. My awards in several national competitions demonstrate my ability to design, prototype and present innovative solutions.

PROFESSIONAL EXPERIENCE

Engineering Intern – SCADA Supervision of an MT Substation
OCP Group – Maroc Chimie Division



📅 Jul–present 2026

📍 Safi Industrial Platform

- Addressed the lack of supervision between the Medium Voltage substation and the control room to secure seawater pumping and workshop power supply.
- Designed automation and remote management architecture to centralize control of 12 MV cells (9 motors/pumps, electrical feeds).
- Configured Sepam Series 80 relays, programmed Modbus TCP/IP on Modicon M580, and developed SCADA under Ignition (alarms, remote control, I/U/P monitoring).

Technical Intern – Electrical Maintenance & Instrumentation
Ciments du Maroc – Heidelberg Materials



📅 Aug–Sep 2025

📍 Safi Plant

- Performed preventive and corrective maintenance operations (crushers, compressors, pumps) with site GMAO tracking.
- Calibrated and diagnosed process instrumentation: gas analyzers, PT100 sensors, pressure, vibration and proximity sensors.
- Conducted technical study of the electrolytic starter (liquid rheostat) for the ball mill's slip-ring asynchronous motor.

Engineering Trainee – Port Crane Reliability Improvement
Marsa Maroc



📅 July 2025

📍 Safi Port

- Conducted reliability analysis of the G40T crane through failure history analysis and evaluation of FMEA, MTBF, MTTR and availability indicators.
- Automated preventive lubrication without human intervention: pump control via hour meter according to manufacturer data.
- Designed a remote IoT solution (Teltonika 4G router) and developed a SCADA HMI for temporal monitoring and lubrication control.

ENGINEERING PROJECTS

Quadcopter Drone for HV Line Inspection

ArduPilot | YOLO | Python

Design and implementation of an autonomous inspection drone for HV lines with AI-powered anomaly detection.

SkyPharma – Cartesian Stock Management Robot

H-Bot | GRBL | Python

Design of a Cartesian robot with web interface and SQLite stock management.

6 DOF Robotic Manipulator Arm

SolidWorks | STM32 | C++ | Python

Mechanical design and embedded control of a 6-axis arm with Python HMI.

Multi-robot Mobile Platform with ADAS Features

ESP-NOW

Development of communicating autonomous robots with PID control and obstacle detection.

7-inch FPV Freestyle Drone

Betaflight | FPV

Design and assembly of a 7-inch FPV drone with Betaflight configuration for freestyle flying.

LEADERSHIP & ASSOCIATION

President – RoboThings Club, FST Mohammedia

2024–2026

- Management, supervision and coordination of club activities.
- Founder of the national competition RoboDays.

Digital Portfolio



Scan to view my online portfolio